Name:_____

Exam Question Booklets
OCR Physics A

Booklet 15
15 Questions
140 marks

Things to remember:

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1

(a) Explain what is meant by the *principle of superposition* of two waves.

.....

.....

..... [2]

(b) Fig. 5.1 shows the arrangement for viewing a visible interference pattern on a screen.

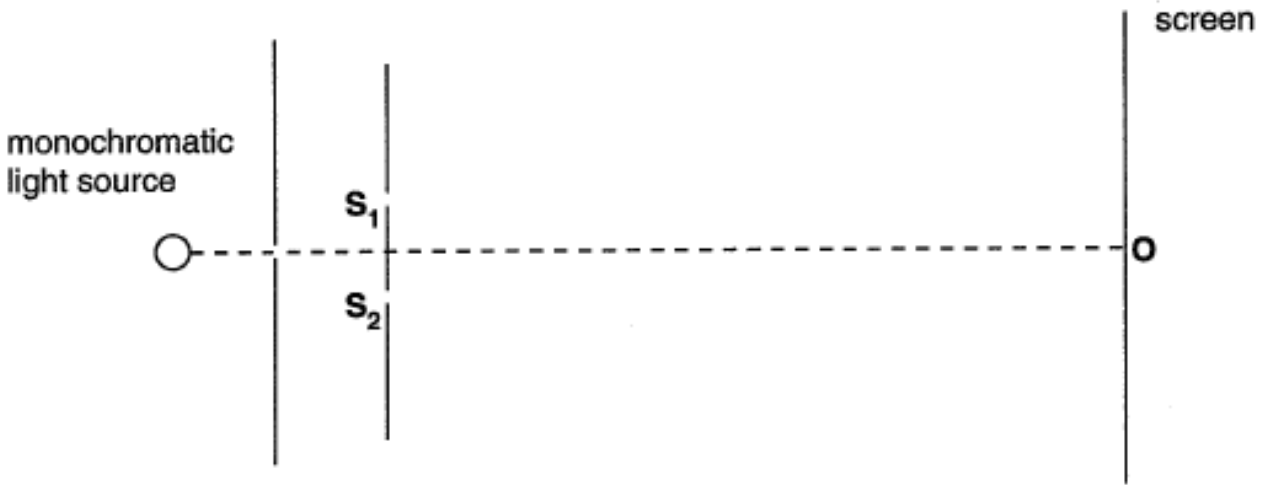

Fig. 5.1

In a darkened room, a double slit (S_1S_2) is placed in front of a narrow single slit situated in front of a monochromatic (one frequency only) light source.

(i) In order to produce a clear interference pattern on the screen, the wave sources must be *coherent*. State what is meant by *coherent*.

.....

..... [1]

(ii) Explain how the arrangement shown ensures that the slits S_1 and S_2 act as coherent light sources.

.....

.....

..... [2]

(iii) The point **O** on the screen is directly opposite the centre of the double slit. State and explain the nature of the interference that occurs at **O**.

.....

.....

..... [2]

(iv) The distance between slits $\mathbf{S_1}$ and $\mathbf{S_2}$ is 0.6 mm. When the screen is placed 1.8 m from the slits, the distance between neighbouring minima in the interference pattern formed on the screen is 2.0 mm. Calculate the wavelength of the light.

wavelength = ... m [3]

(v) State and explain how the interference pattern changes when light of a shorter wavelength is used in the experiment.

.....

.....

..... [2]

2

Explain why electrons can be emitted from a clean metal surface illuminated with bright ultra-violet light but never when infra-red light is used, however intense.

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..... [2]

3

(a) A student solders two resistors together as shown in Fig. 5.1.

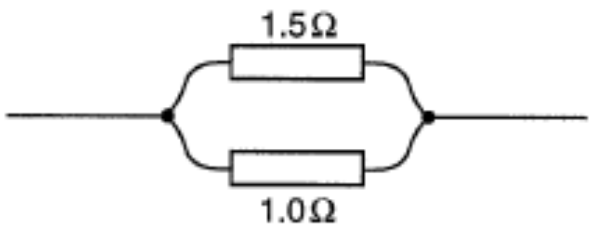

Fig. 5.1

(i) State whether the two resistors are connected in a series or in a parallel combination.

..... [1]

(ii) Show that the total resistance of the combination of resistors is 0.6 Ω.

[2]

(b) Fig. 5.2 shows the soldered resistors from **(a)** connected across the terminals of a cell.

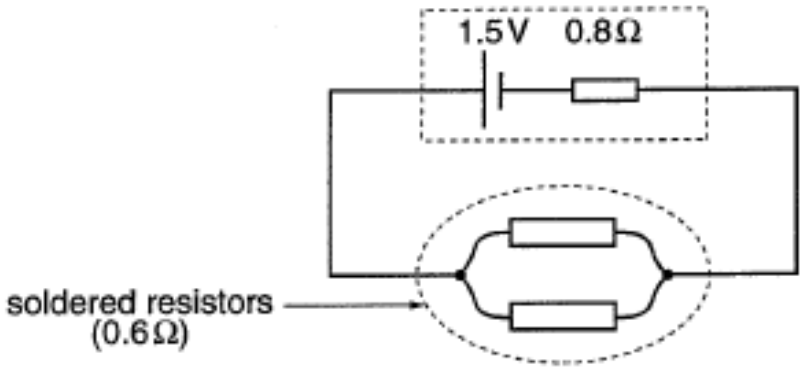

Fig. 5.2

The cell has internal resistance 0.8 Ω and e.m.f. 1.5 V.

(i) Define *e.m.f.* in terms of energy transformed and electric charge.

.....

.....[2]

(ii) Suggest why a cell has internal resistance.

..... [1]

(iii) Calculate the total resistance R of the circuit in Fig. 5.2.

R = ... Ω [2]

(iv) Hence calculate the current I in the circuit.

I = ... A [2]

(v) 1. Write an equation for the power dissipated by a current-carrying resistor.

2. For the circuit in Fig. 5.2, calculate the ratio:

$$\frac{\text{power dissipated by internal resistance}}{\text{power dissipated by total external resistance}}.$$

ratio = ... [4]

4

Fig. 4.1 shows a car battery of e.m.f. 12 V and internal resistance 0.014 Ω connected to the starter motor of a car. When the car engine is being started, the car battery provides a current of 160 A to the starter motor.

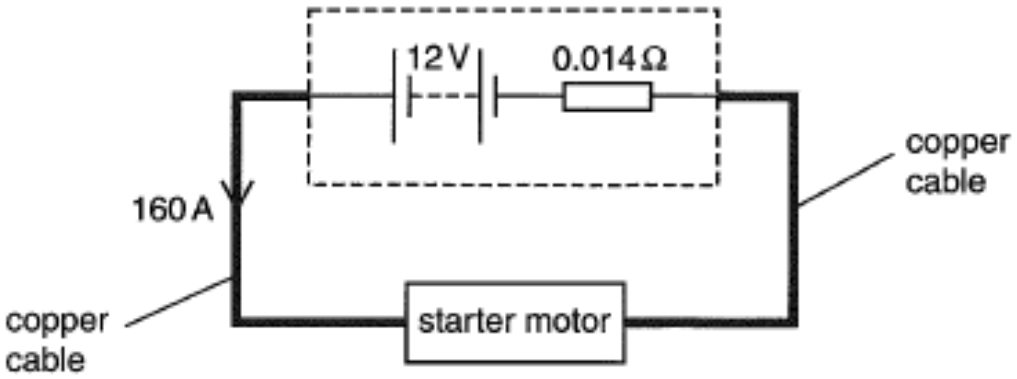

Fig. 4.1

(a) Show that the p.d. across the internal resistance is about 2.2 V.

[1]

(b) Determine the terminal p.d. across the battery.

p.d. = ... V [1]

(c) The cables connecting the battery to the starter motor have total length 0.85 m and diameter 8.0 mm.

(i) Show that the cross-sectional area of the cable is 5.0×10^{-5} m^2.

[1]

(ii) The cables are made from copper of resistivity 1.7×10^{-8} Ω m. Calculate the total resistance of the cables.

resistance = ... Ω [3]

(iii) State and explain how your answer to **(c)(ii)** would change if the cable had half the length but twice the diameter.

.....

.....

.....

.....[2]

5

(a) Complete the following sentence for a statement of Kirchhoff's first law.

The sum of the ..... into a point in a circuit is equal to the sum of the ..... out from that point. [1]

(b) Complete the following statement about Kirchhoff's second law.

In an electrical circuit, the sum of the e.m.f.s around a closed loop is equal to the sum of the p.d.s around the same loop. This is a consequence of conservation of ..... . [1]

(c) Fig. 5.1 shows a part of an electrical circuit.

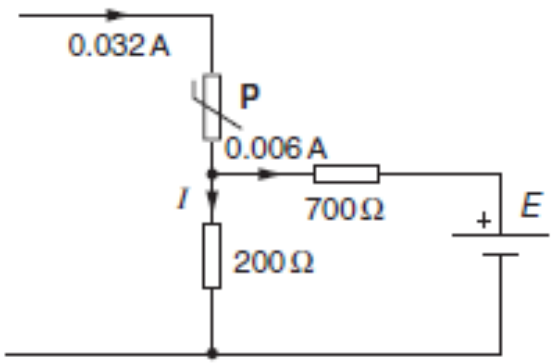

Fig. 5.1

(i) Name the component **P**.

.....[1]

(ii) State how the resistance of component **P** is affected by a change in its temperature.

.....

.....[1]

(iii) At a particular temperature, the currents are as shown in Fig. 5.1. Use Kirchhoff's laws to determine

1 the current I in the 200 Ω resistor

I = ... A [1]

2 the e.m.f. E of the cell.

E = ... V [3]

A phenomenon associated with microwave ovens is the uneven heating of food. An internet website gives the following explanation and the illustration shown in Fig. 5.1.

> 'Microwaves of a fixed frequency are emitted in all directions from a source within the oven. The waves reflect off the metal walls so that the microwave radiation reaching any particular point arrives both directly and by reflection. The waves interfere and set up standing waves. This produces the pattern of hot and cold zones observed in food heated in the oven.'

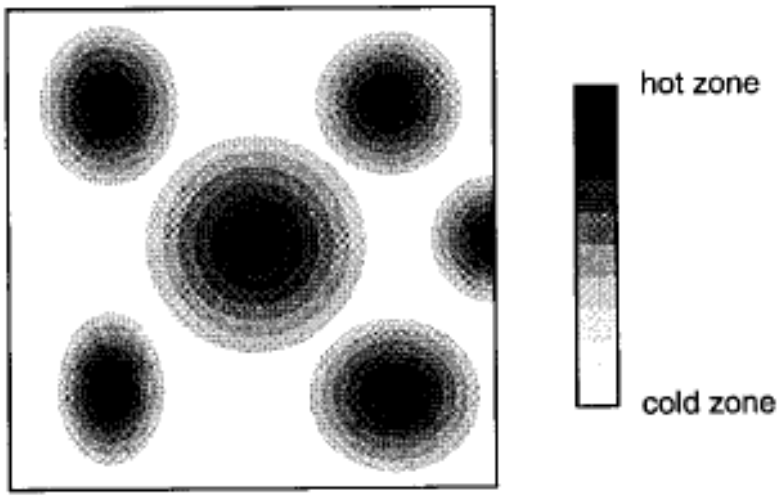

Fig. 5.1

(a) State how the reflected microwaves set up standing (stationary) waves in the oven.

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.....

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.....[2]

(b) Mark on Fig. 5.1 the positions of two antinodes – label these as **A**. [1]

(c) The frequency of the microwaves is 2.45×10^9 Hz. Calculate the wavelength of the microwaves.

wavelength = ...m [3]

(a) When used to describe stationary (standing) waves explain the terms

(i) node .....

..... **[1]**

(ii) antinode. .....

..... **[1]**

(b) Fig. 5.1 shows a string fixed at one end under tension. The frequency of the mechanical oscillator close to the fixed end is varied until a stationary wave is formed on the string.

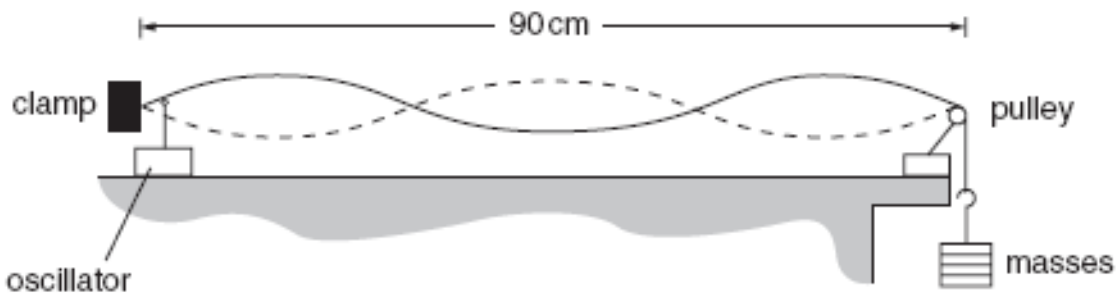

Fig. 5.1

(i) Explain with reference to a progressive wave on the string how the stationary wave is formed.

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.....

..... **[3]**

(ii) On Fig. 5.1 label one node with the letter **N** and one antinode with the letter **A**. **[1]**

(iii) State the number of antinodes on the string in Fig. 5.1.

number of antinodes = ... **[1]**

(iv) The frequency of the oscillator causing the stationary wave shown in Fig. 5.1 is 120 Hz.

The length of the string between the fixed end and the pulley is 90 cm.

Calculate the speed of the progressive wave on the string.

speed = ... $m\,s^{-1}$ **[3]**

(c) The speed v of a progressive wave on a stretched string is given by the formula

$$v = k\sqrt{W}$$

where k is a constant for that string. W is the tension in the string which is equal to the weight of the mass hanging from the end of the string.

In **(b)** the weight of the mass on the end of the string is 4.0 N. The oscillator continues to vibrate the string at 120 Hz. Explain whether or not you would expect to observe a stationary wave on the string when the weight of the suspended mass is changed to 9.0 N.

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..... **[3]**

8

Fig. 4.1 shows a tall hollow cylinder with one open end. A standing wave can be formed in the air column in the cylinder by holding a vibrating tuning fork near the top.

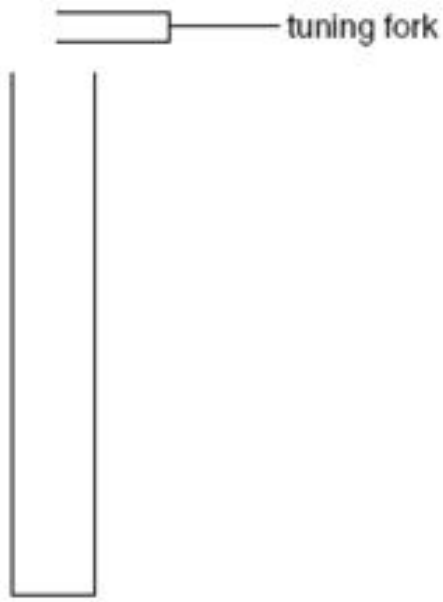

Fig. 4.1

(i) The air column in Fig. 4.1 is producing its lowest frequency note (fundamental). Label on Fig. 4.1 the positions in the cylinder of a node, with the letter **N**, and an antinode with the letter **A**. **[2]**

(ii) The length of the air column is 0.64 m and the speed of sound in air is $330\,m\,s^{-1}$. Calculate the frequency of this standing wave.

frequency = ...Hz **[3]**

Fig. 4.2 shows a cylinder identical to that shown in Fig. 4.1 except that it has two open ends. A standing wave is again formed in the air column in the cylinder by another vibrating tuning fork. The standing wave is vibrating in its fundamental mode.

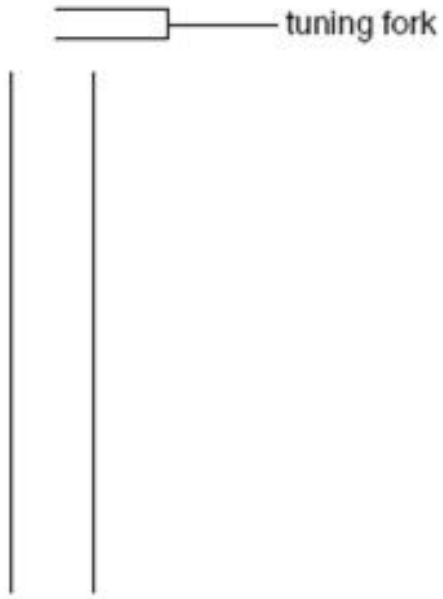

Fig. 4.2

(i) Mark the positions of a node **N** and antinodes **A** in the air column. **[1]**

(ii) Determine the frequency of this standing wave.

frequency = ...Hz **[1]**

9

In this question, one mark is available for the quality of written communication.

Draw a circuit diagram to show how a light-dependent resistor, a voltmeter, a variable resistor and a cell may be used as a potential divider circuit to monitor changes in the light intensity in the laboratory. Explain the operation of your circuit and suggest a reason for using a variable resistor.

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[9]

(a) Fig. 3.1 shows the circuit symbol for a particular component.

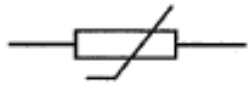

Fig. 3.1

Name the component and state how its resistance changes as the temperature of the component is increased.

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..... [2]

(b) Fig. 3.2 shows a potential-divider circuit. The battery has negligible internal resistance and the voltmeter has very high resistance.

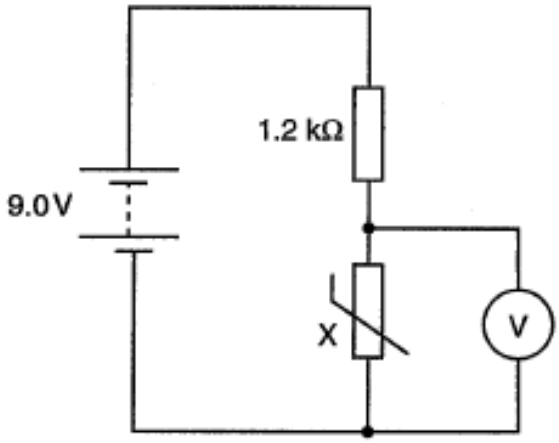

Fig. 3.2

(i) At a particular temperature, the resistance of component X is 4.2 kΩ. Calculate the voltmeter reading.

voltmeter reading = ... V [3]

(ii) State how your answer to **(b)(i)** changes when the temperature of the component X is **increased**.

.....

..... [1]

Fig. 8.1 shows an isolated zinc plate exposed to weak ultra violet (u.v.) light.

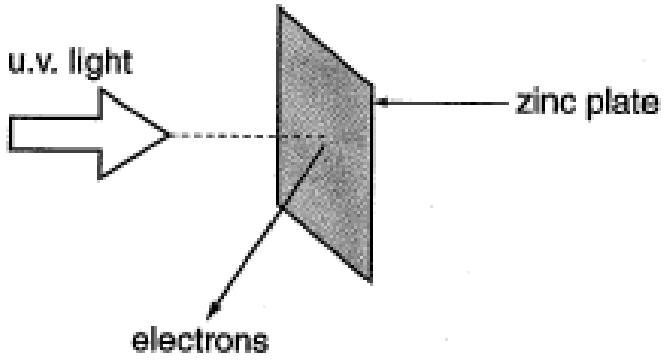

Fig. 8.1

The u.v. light causes electrons to be emitted from the surface of the plate.

(a) Name this phenomenon.

..... [1]

(b) Initially, the plate is neutral in charge. State and explain the effect on the charge of the plate as the zinc plate is exposed to the u.v. light.

.....

.....

.....

..... [2]

(c) State and explain the effect on the rate of emission of electrons when the intensity of the u.v. light is **increased**.

.....

.....

.....

..... [2]

(d) In a databook, the *work function energy* of zinc is quoted as 4.24 eV. Explain what is meant by work function energy (no calculations are necessary).

.....

..... [1]

(a) In an experiment, an electrically insulated zinc plate is negatively charged. When exposed to a weak ultra-violet source, the plate starts to lose its negative charge.

Explain this phenomenon in terms of the photoelectric effect. Suggest how increasing the intensity of the ultra-violet source would affect the experiment.

(You will be awarded marks for the quality of written communication in your answer to this question.)

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..... [7]

(b) A 1.0 mW laser produces red light of wavelength 6.3×10^{-7} m.

(i) Calculate

1. the frequency of the radiation,

frequency = ... unit ..... [3]

2. the energy of a photon of red light.

energy = ... J [2]

(ii) Calculate the number of photons emitted per second by the laser.

number = ... s^{-1} [2]

(iii) State how, and explain why, the number of photons emitted per second would change if the 1.0 mW laser produced blue light.

.....

.....

..... [2]

(a) Einstein's photoelectric equation may be written as

$$hf = \phi + \tfrac{1}{2}mv_{max}^2$$

Identify the terms

hf .....

ϕ .....

$\tfrac{1}{2}mv_{max}^2$ ..... [3]

(b) The surface of sodium metal is exposed to electromagnetic radiation of wavelength 6.5×10^{-7} m. This wavelength is the maximum for which photoelectrons are released.

(i) Calculate the threshold frequency.

frequency = ... unit: ..... [3]

(ii) Show that the work function energy of the metal is 1.9 eV.

[3]

(c) For a particular wavelength of incident light, sodium releases photoelectrons. State how the rate of release of photoelectrons changes when the intensity of light is doubled. Explain your answer.

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..... [2]

Fig. 7.1 shows an electrical circuit, which includes a photocell.

14

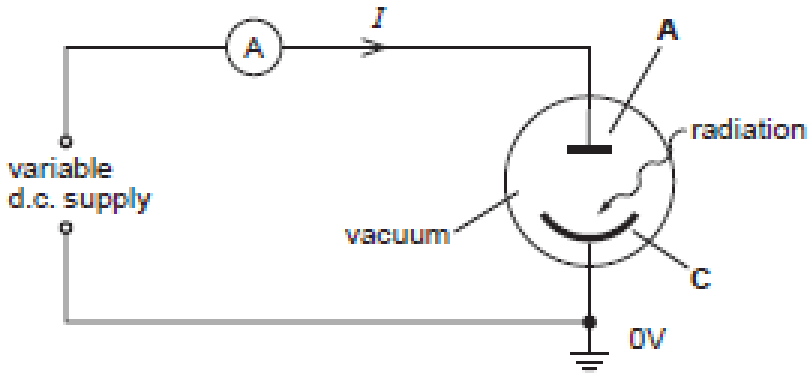

Fig. 7.1

The photocell consists of a metal plate **C** that is exposed to electromagnetic radiation. The photoelectrons emitted travel towards the electrode **A**. A sensitive ammeter measures the current in the circuit.

The plate **C** is illuminated with ultraviolet radiation of constant intensity and of wavelength 2.5×10^{-7} m. Fig. 7.2 shows how the photoelectric current I in the circuit varies with the potential difference V between **A** and **C**.

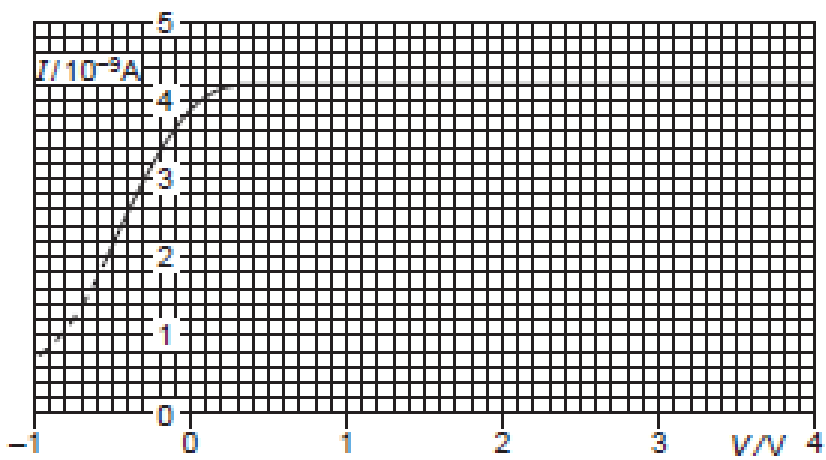

Fig. 7.2

(a) Use Fig. 7.2 to show that when the potential difference V is 2.0V the number of electrons reaching the electrode **A** per second is $2.6 \times 10^{10}\,s^{-1}$.

[3]

(b) The metal of plate C has work function energy 2.2 eV. Calculate the maximum kinetic energy in joules of the emitted photoelectrons from this plate.

kinetic energy = ... J [4]

(c) **(i)** State how the maximum energy of the photoelectrons emitted from plate C depends on the intensity of the incident radiation.

.....

.....[1]

(ii) State and explain how the photoelectric current depends on the intensity of the radiation.

.....

.....

.....

.....[2]

(a) State **one** main property of an electromagnetic wave.

.....

.....[1]

(b) Arrange the following electromagnetic radiations in the order of increasing wavelengths. Start with the **shortest** wavelength.

infrared visible light gamma rays microwaves

.....[1]

(c) The table below shows the work function energies of some metals.

metal	work function energy (eV)
beryllium	5.0
magnesium	3.7
potassium	2.3
silver	4.7
zinc	4.3

(i) Define the *work function energy* of a metal.

.....

.....[1]

(ii) State and explain which metal has the **lowest** threshold frequency.

.....

.....

.....[2]

(iii) A plate made of magnesium is illuminated with electromagnetic waves of wavelength 3.2×10^{-7} m. The electrons emitted from the surface of the plate have a range of kinetic energies.

1 Describe how the incident radiation interacts with the metal to release electrons from its surface and explain why these electrons are emitted with a range of kinetic energies.

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.....[4]

2 Show that the maximum kinetic energy of the electrons emitted from the surface of the magnesium plate is 3.0×10^{-20} J.

[3]